Functions and gradient functions

1

a $x = 4$

b $(-\infty, 4) \cup (4, \infty)$

2

a Zero

b Positive

c Negative

3

a

i $x = -2$

ii $(-\infty, -2)$

iii $(-2, \infty)$

b

i $x = -1$

ii $(-1, \infty)$

iii $(-\infty, -1)$

c

i $x = 1, -1$

ii $(-\infty, -1) \cup (1, \infty)$

iii $(-1, 1)$

d

i $x - 1, 1, 3$

ii $(-\infty, -1) \cup (1, 3)$

iii $(-1, 1) \cup (3, \infty)$

e

i $-6, 0$

ii $(-\infty, -6) \cup (0, \infty)$

iii $(-6, 0)$

f

i $x = 2, -2, -3$

ii $(-3, -2) \cup (2, \infty)$

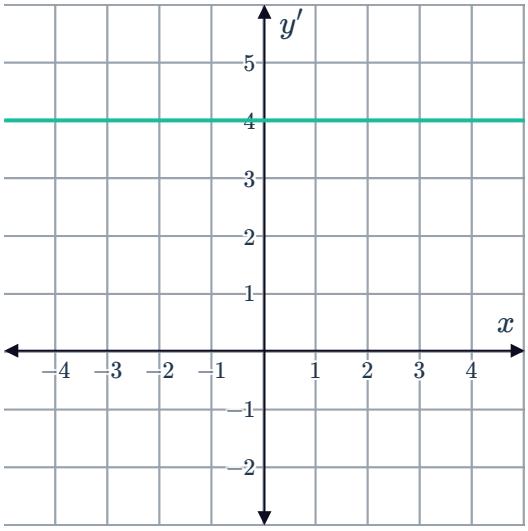
iii $(-\infty, -3) \cup (-2, 2)$

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4

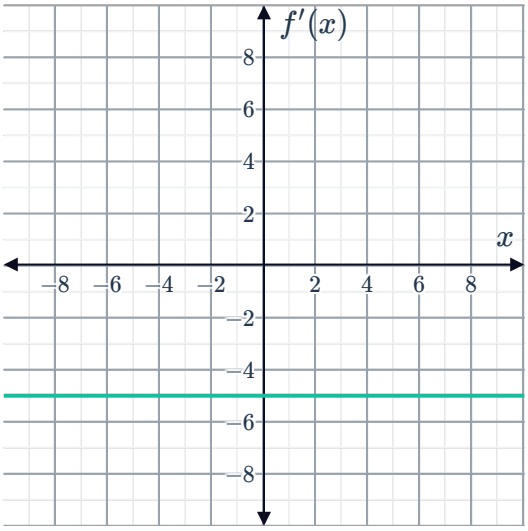
a $\dfrac{dy}{dx} = 4$

b

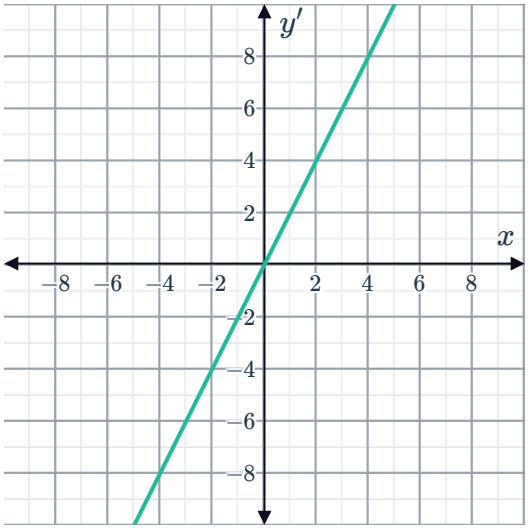


5

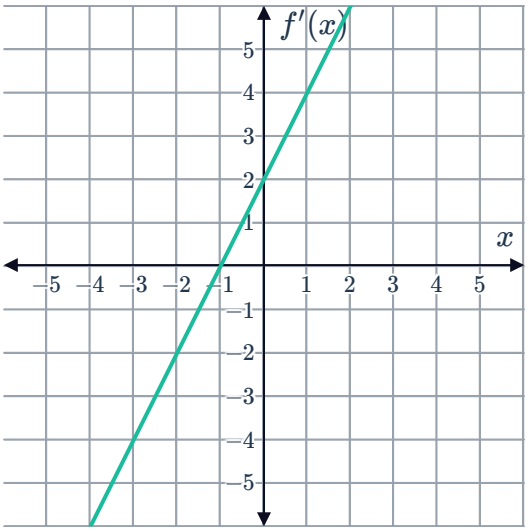
a



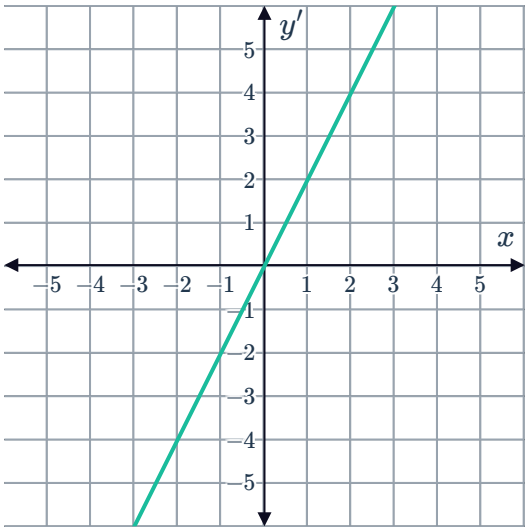
b



c

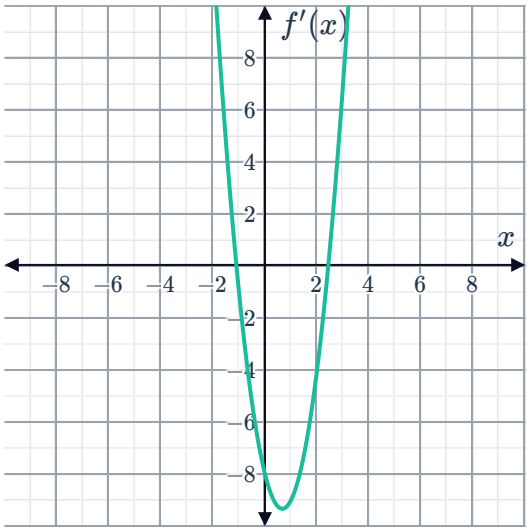


d



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e



6

a $x = 5$ **b** Below **c** Above

7

a $x = -7$ **b** Above **c** Below

8

a $x = -5$ **b** Above **c** Below

9

a $x = -5$ **b** Below **c** Above

10

a $(5, 3)$ **b** 0

c The x-intercept. **d** Above

e Above **f** Minimum turning point

11

a $(-7, -3)$ **b** 0

c The x-intercept **d** Below

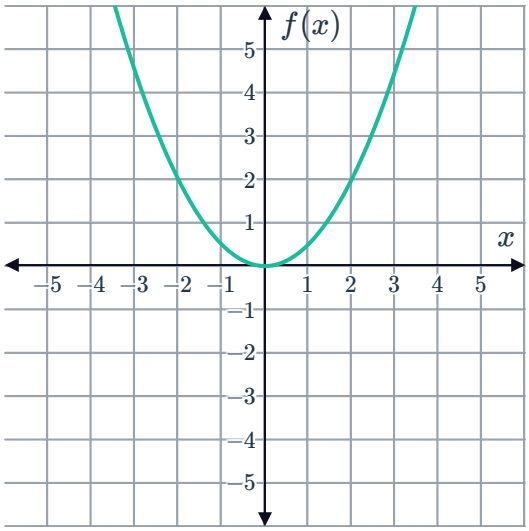
e Below **f** Maximum turning point

12 $f(x)$ is a parabola with a minimum turning point at $x = 4$.

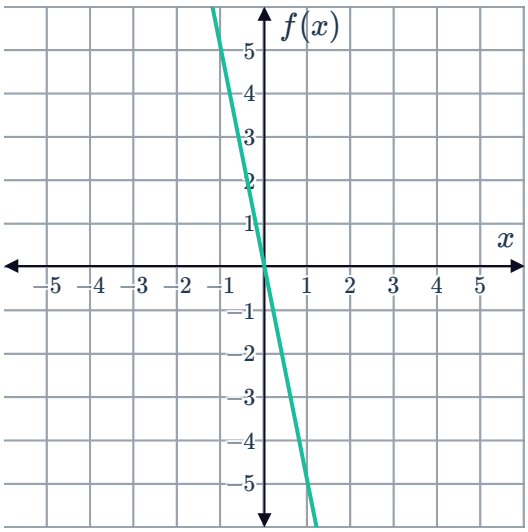
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13

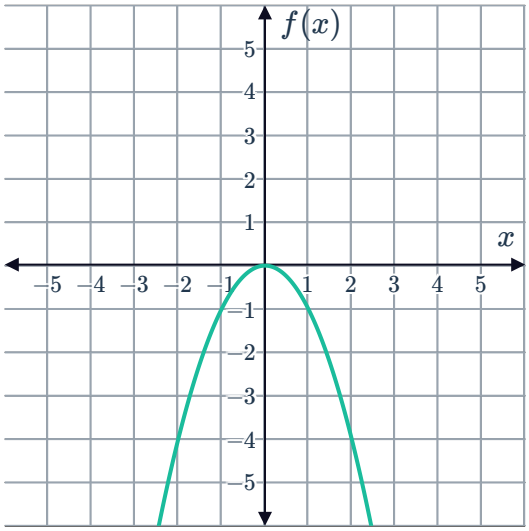
a A concave up parabola with vertex at $x = 0$:



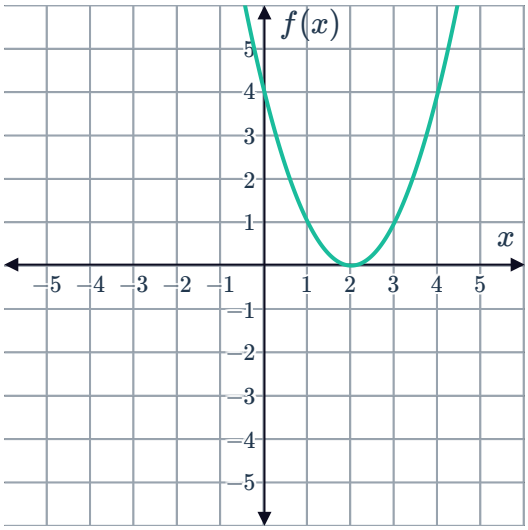
b A line with gradient of -5:



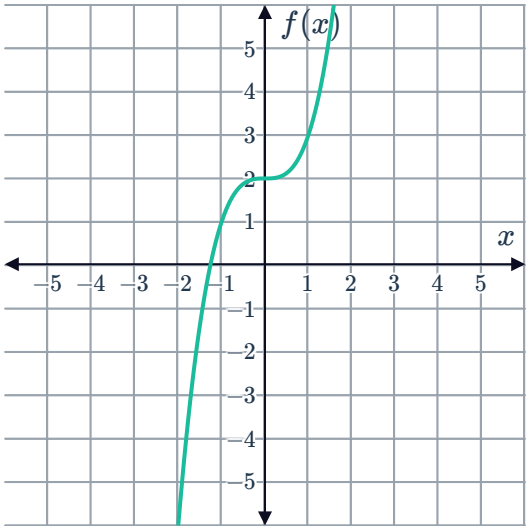
c A concave down parabola with vertex at $x = 0$:



d A concave up parabola with vertex at $x = 2$:



e An increasing cubic function with horizontal point of inflection at $x = 0$:



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Types of stationary points

14

a Minimum turning point

b Maximum turning point

c Horizontal point of inflection

15

a

i $\frac{dy}{dx} = -12x + 84$

ii $(7, 265)$

iii $(7, 265)$ is a maximum turning point

b

i $\frac{dy}{dx} = 3x^2 - 42x + 144$

ii $(6, 305), (8, 301)$

iii $(6, 305)$ is a local maximum, and $(8, 301)$ is a local minimum

c

i $f'(x) = 3x^2 + 24x + 45$

ii $(-5, 4), (-3, 0)$

iii $(-5, 4)$ is a local maximum, and $(-3, 0)$ is local a minimum

d

i $f'(x) = 3x^2 + 30x + 75$

ii $(-5, 4)$

iii $(-5, 4)$ is a horizontal point of inflection

e

i $f'(x) = -x^2 + 13x - 30$

ii $\left(3, -\frac{61}{2}\right), \left(10, \frac{80}{3}\right)$

iii $\left(3, -\frac{61}{2}\right)$ is a local minimum, and $\left(10, \frac{80}{3}\right)$ is a local maximum

f

i $f'(x) = 8x + 9$

ii $\left(-\frac{9}{8}, -\frac{1}{16}\right)$

iii $\left(-\frac{9}{8}, -\frac{1}{16}\right)$ is a minimum turning point

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g

i $f'(x) = -300 + 480x - 192x^2$

ii $\left(\frac{5}{4}, 9\right)$

iii $\left(\frac{5}{4}, 9\right)$ is a horizontal point of inflection.

h

i $f'(x) = 4x^3 - 36x$

ii $(3, 4), (-3, 4), (0, 85)$

iii $(3, 4)$ and $(-3, 4)$ are minimum turning points and $(0, 85)$ is a local maximum

16

a $\left(\frac{1}{2}, \frac{21}{4}\right)$

b 0

c Minimum turning point

17

a $f'(x) = 2x + 4$

b $x > -2$

c $x < -2$

d $(-2, 5)$

e Maximum turning point

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18

a $f'(x) = 12x^2 + 10x$

b $\left(-\frac{5}{6}, -\frac{307}{108}\right), (0, -4)$

c

x	-2	$-\frac{5}{6}$	$-\frac{1}{2}$	0	1
$f'(x)$	28	0	-2	0	22

d

i $(0, -4)$ **ii** $\left(-\frac{5}{6}, -\frac{307}{108}\right)$

e No, because the function is a cubic so it approaches $-\infty$.

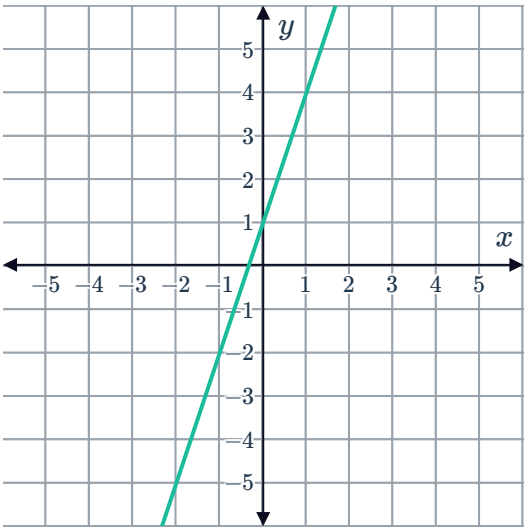
19

a $f'(x) = 6x - 54$ **b** $x = 9$

c Minimum turning point

Graphs of functions

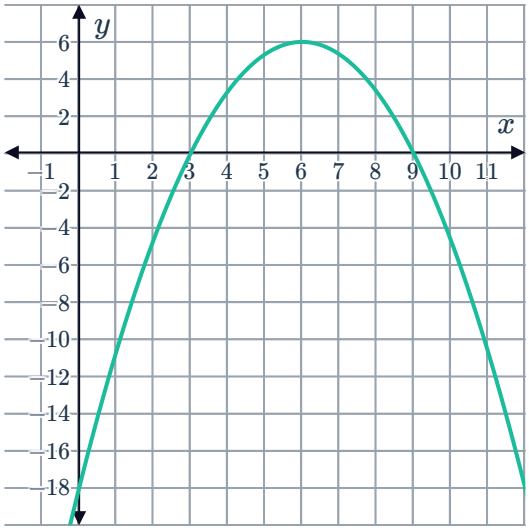
20



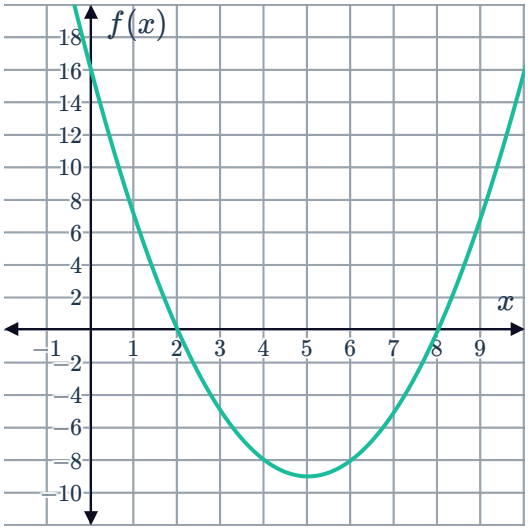
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21

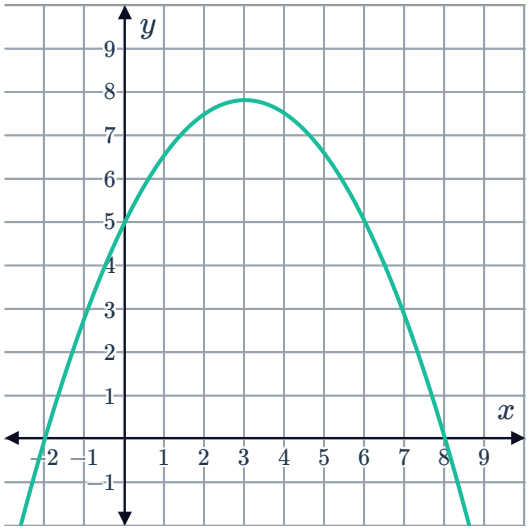
a



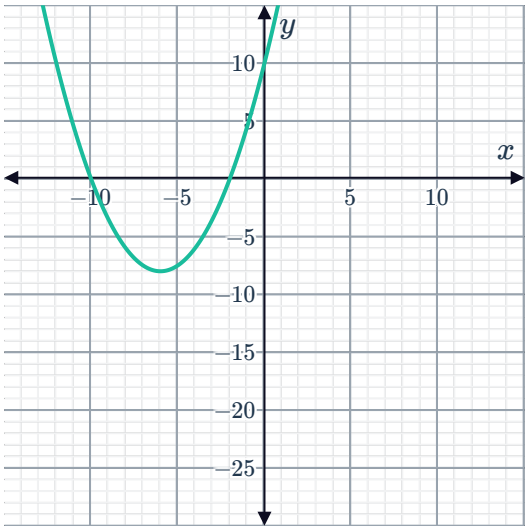
b



c

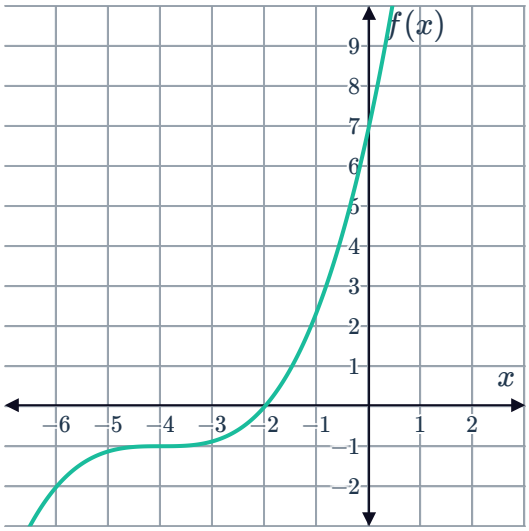


d

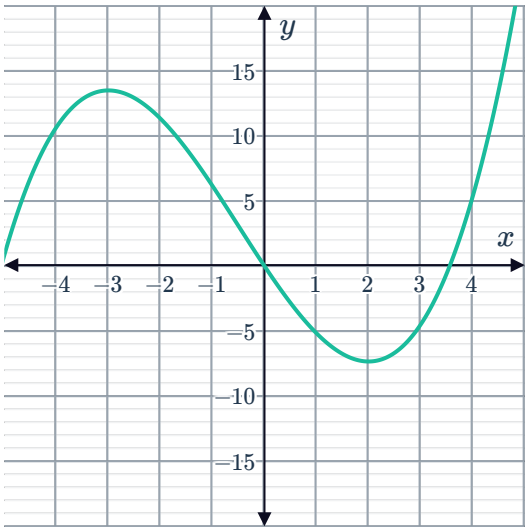


22

a



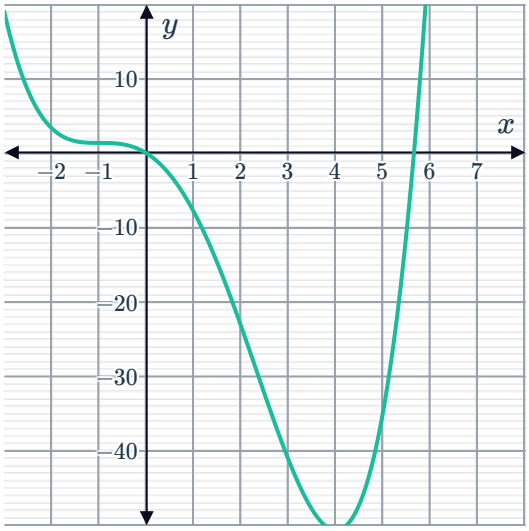
b



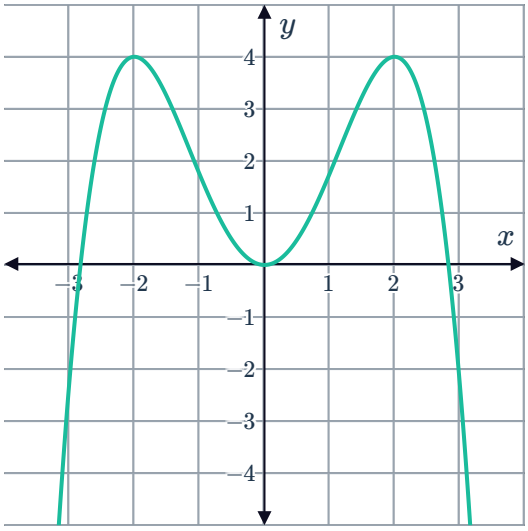
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23

a



b



24

a $x = 2, 4$

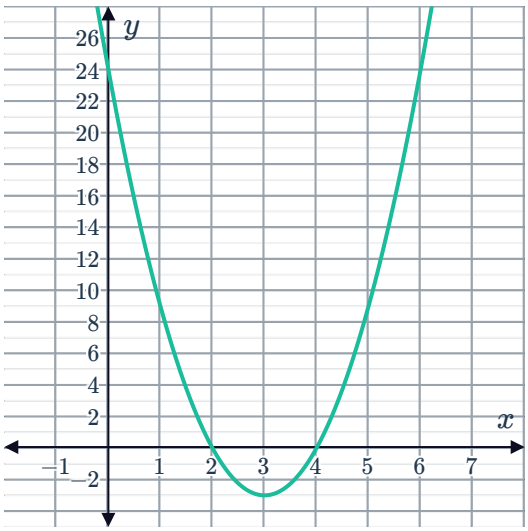
b $y = 24$

c $6x - 18$

d $(3, -3)$

e $(3, -3)$ is a minimum turning point

f



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25

a

i $y = -16$

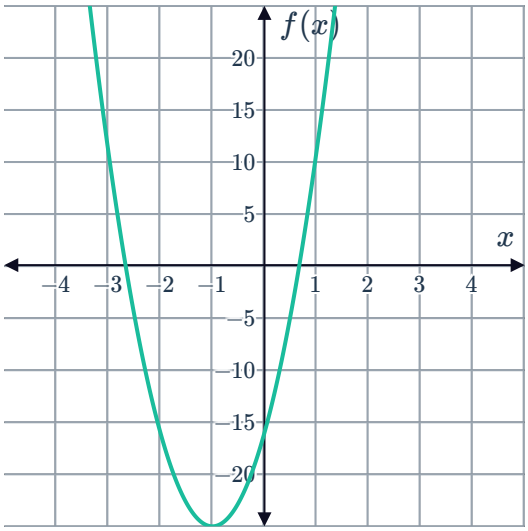
ii $x = \frac{2}{3}, -\frac{8}{3}$

iii $f'(x) = 18x + 18$

iv $x = -1$

v Minimum turning point

vi



b

i $y = -25$

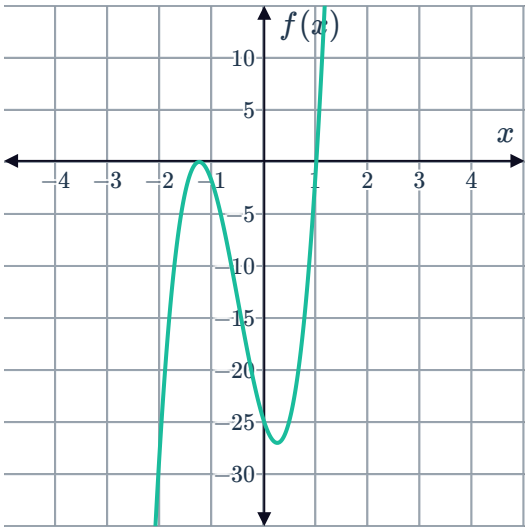
ii $x = -\frac{5}{4}, 1$

iii $f'(x) = 3(4x + 5)(4x - 1)$

iv $x = -\frac{5}{4}, \frac{1}{4}$

v $\left(\frac{1}{4}, -27\right)$ is a local minimum, $\left(-\frac{5}{4}, 0\right)$ is a local maximum.

vi



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c

i $y = 1$

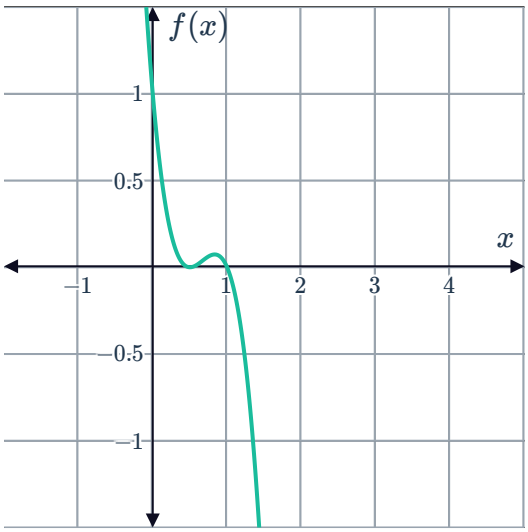
ii $x = \frac{1}{2}, 1$

iii $f'(x) = (2x - 1)(5 - 6x)$

iv $x = \frac{1}{2}, \frac{5}{6}$

v $\left(\frac{5}{6}, \frac{2}{27}\right)$ is a local maximum, $\left(\frac{1}{2}, 0\right)$ is a local minimum.

vi



d

i $y = 7$

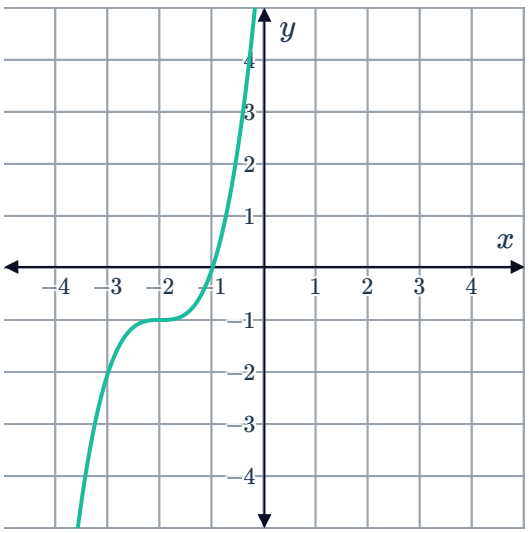
ii $x = -1$

iii $f'(x) = 3(x + 2)^2$

iv $x = -2$

v Horizontal point of inflection

vi



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e

i $y = 0$

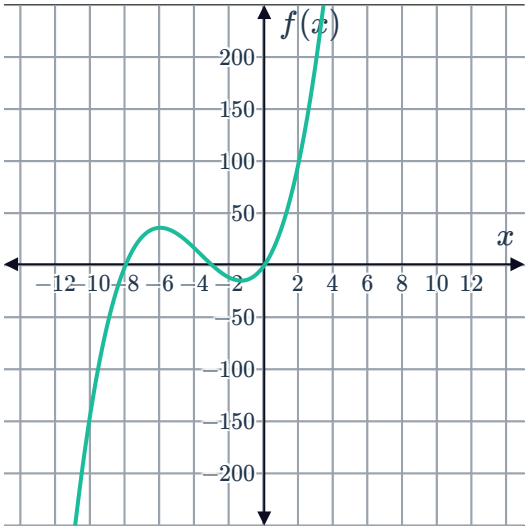
ii $x = 0, -8, -3$

iii $f'(x) = 3x^2 + 22x + 24$

iv $x = -6, -\frac{4}{3}$

v $(-6, 36)$ is a local maximum, $\left(-\frac{4}{3}, -\frac{400}{27}\right)$ is a local minimum.

vi



f

i $y = 20$

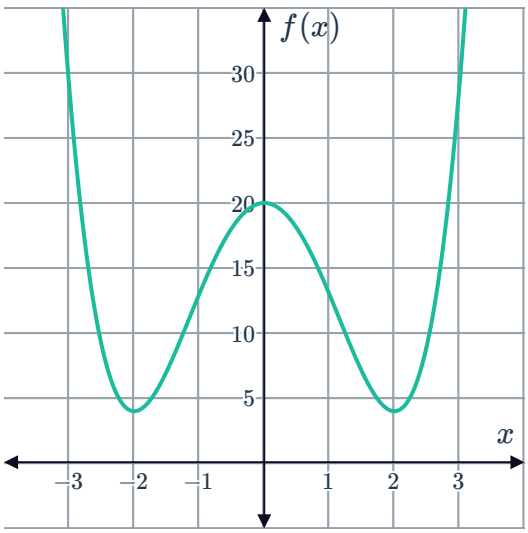
ii $x = \pm 2$

iii $f'(x) = 4x\left(x^2 - 4\right)$

iv $x = 0, 2, -2$

v $(0, 20)$ is a local maximum , $(-2, 4)$ and $(2, 4)$ are minimum turning points.

vi



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g

i $y = -18$

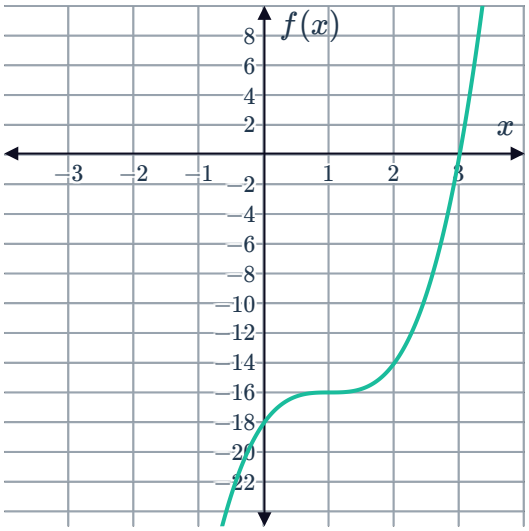
ii $x = 3$

iii $f'(x) = 6(x-1)^2$

iv $x = 1$

v Horizontal point of inflection.

vi



h

i $y = 0$

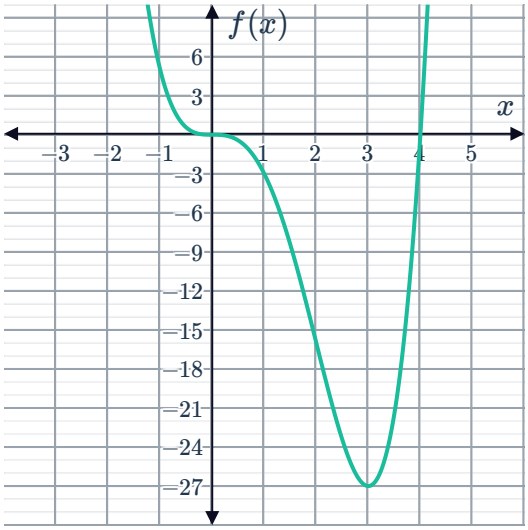
ii $x = 0, 4$

iii $f'(x) = 4x^3 - 12x^2$

iv $x = 0, 3$

v $(0, 0)$ is a horizontal point of inflection, and $(3, -27)$ is a minimum turning point.

vi



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i

i $y = 0$

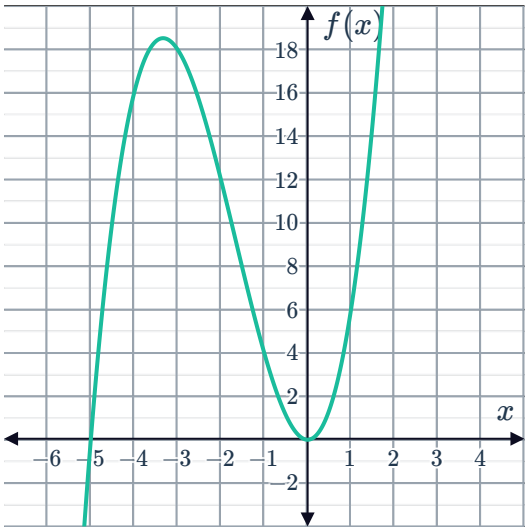
ii $x = 0, -5$

iii $f'(x) = 3x^2 + 10x$

iv $x = 0, -\frac{10}{3}$

v $(0, 0)$ is a local minimum, and $(-\frac{10}{3}, \frac{500}{27})$ is a local maximum.

vi



j

i $y = 0$

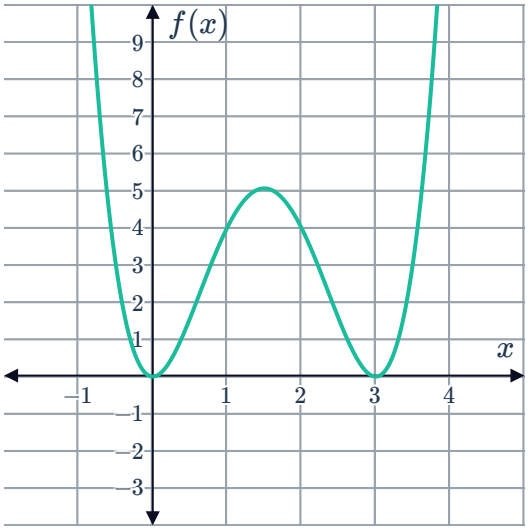
ii $x = 0, 3$

iii $f'(x) = 2x(x-3)(2x-3)$

iv $x = 0, 3, 1.5$

v $(0, 0)$ and $(3, 0)$ are minimum turning points, and $(1.5, 5)$ is a local maximum.

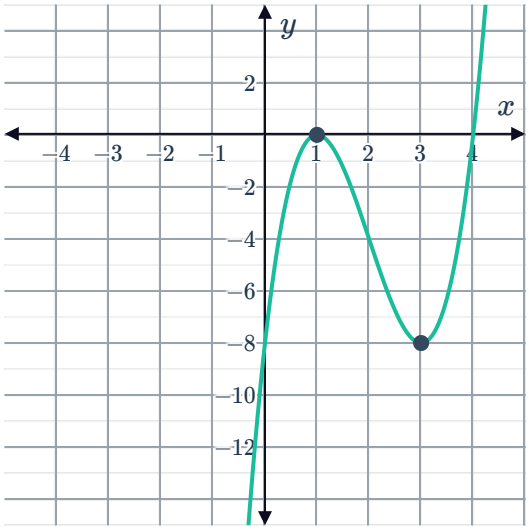
vi



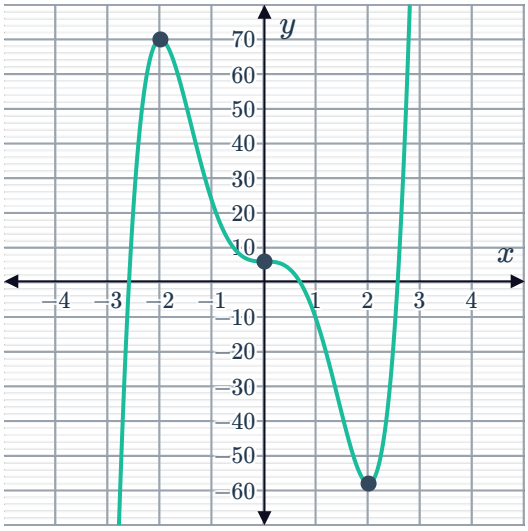
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26

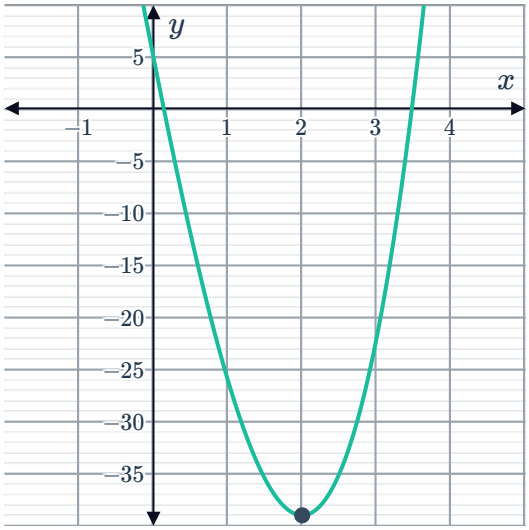
a



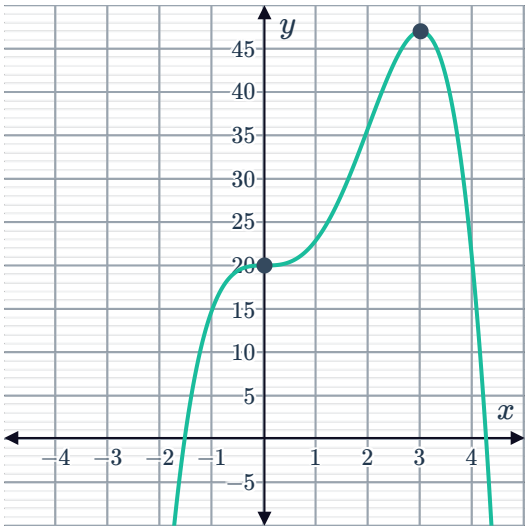
b



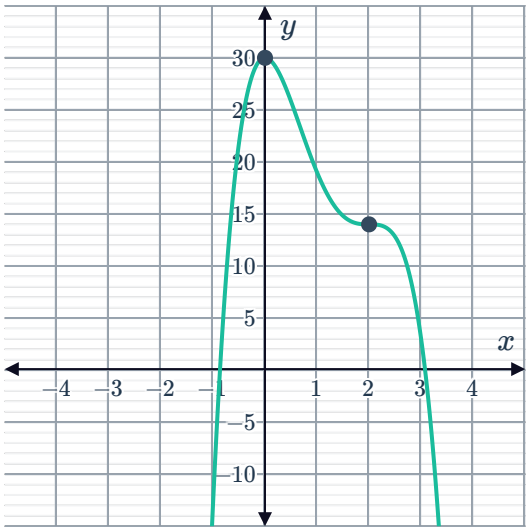
c



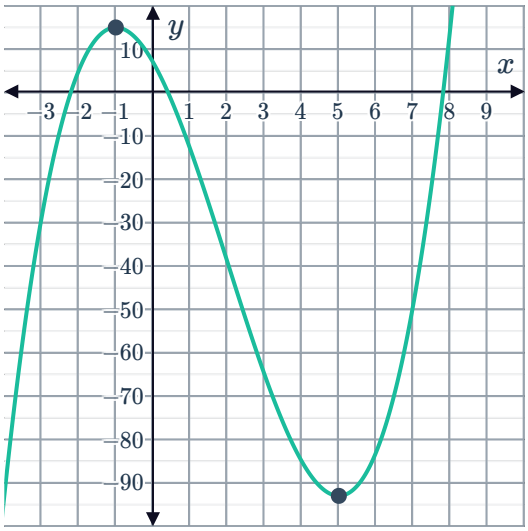
d



e



f



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